

SIMATS ENGINEERING

ASSESSMENT TOOL III – ERROR ANALYSIS TASK (LOW)

Course Name:	Code:	CO: CO2
Duration: 90 min Date:	Total Marks:	Reg. No / Name : 192325076 / SHAM S.

CO2: Execute footprinting in social engineering such as DNS analysis, competitive intelligence gathering, and attack simulations to collect relevant information for identifying potential security risks.(BL3)

Assessment Tool Weightage: 35%

Instructions: Answer all questions. Each question carries 10 marks.

1. A student performs footprinting using Google dorks but constructs queries without operators like “site:”, “filetype:”, or quotation marks. As a result, the search results include unrelated websites and irrelevant documents. The student proceeds with analysis using this incorrect data. Identify the error in query formulation and explain why improper search syntax leads to inaccurate footprinting. Demonstrate how to correct the search query using proper operators and refine results for accurate intelligence gathering.

The student's mistake is the incorrect formulation of Google search queries by not using Google Dork operators such as site:, filetype:, intitle:, inurl:, or quotation marks (""). Without these operators, Google performs a broad search and returns many unrelated or irrelevant results, making the footprinting process inaccurate and unreliable.

Improper search syntax leads to inaccurate footprinting because:

- It retrieves unrelated websites and documents.
- It increases false-positive results.
- It wastes time analyzing irrelevant information.
- It may cause important target-specific information to be overlooked.
- It reduces the accuracy of reconnaissance and intelligence gathering.

To obtain accurate results, the search query should be refined using appropriate Google Dork operators.

Example:

Incorrect Query:

employee email company reports

This returns results from many unrelated websites.

Correct Query:

site:example.com "employee email" filetype:pdf

or

site:example.com filetype:pdf "annual report"

Explanation of Operators:

- **site:** – Limits results to a specific website or domain.
- **filetype:** – Searches for a particular file type such as PDF, DOCX, or XLSX.
- **Quotation Marks (")** – Searches for the exact phrase.
- **intitle:** – Finds pages containing specific words in the title.
- **inurl:** – Searches for keywords within the URL.

By using these operators correctly, the search results become more precise, relevant, and useful for accurate footprinting and intelligence gathering.

2. While conducting competitive intelligence, a student gathers company data from blogs and unofficial forums without verifying authenticity. The collected information includes outdated employee details and incorrect organizational structure. Identify the mistake in source validation and explain why unreliable data affects decision-making. Recommend methods to verify information using official sources such as company websites, LinkedIn, and government databases.

The student's mistake is using unverified and unofficial sources such as blogs and online forums without checking whether the information is accurate, current, or trustworthy. Competitive intelligence should always rely on credible and verified sources to ensure the collected information is reliable.

Using unreliable data negatively affects decision-making because:

- It may contain outdated or incorrect information.
- It can lead to inaccurate analysis of the organization.
- It may result in poor business or security decisions.
- It wastes time and resources by acting on false information.
- It reduces the credibility of the intelligence gathered.

To verify the authenticity of information, the following official sources should be used:

- **Company Website:** Verify company profiles, management details, products, services, contact information, and organizational announcements.

- **LinkedIn:** Confirm employee names, job titles, organizational structure, and company updates.
- **Government Databases:** Check company registration details, legal status, business licenses, and regulatory information.
- **Official Press Releases:** Verify recent company announcements and business developments.
- **Annual Reports and SEC/Corporate Filings (where applicable):** Confirm financial information, executive management, and organizational structure.

By collecting information only from **official and trusted sources** and cross-verifying it across multiple platforms, competitive intelligence becomes more accurate, reliable, and useful for informed decision-making.

3. A student directly starts active scanning on a target organization without performing passive footprinting such as WHOIS lookup and search engine analysis. This results in detection alerts being triggered. Identify the mistake in reconnaissance methodology and explain why passive reconnaissance is critical. Suggest the correct sequence of footprinting steps.

The student's mistake is starting active reconnaissance before completing passive footprinting. Active scanning directly interacts with the target's systems, making it more likely to be detected by firewalls, intrusion detection systems (IDS), or intrusion prevention systems (IPS). As a result, security alerts are triggered, revealing the presence of the reconnaissance activity.

Passive reconnaissance is critical because:

- It gathers information without directly interacting with the target's systems.
- It minimizes the risk of detection and alert generation.
- It provides valuable information such as domain names, IP addresses, DNS records, employee details, and organizational information.
- It helps plan targeted and efficient active scanning.
- It reduces unnecessary network traffic and avoids early exposure.

Correct Sequence of Footprinting Steps:

1. **Identify the Target Organization** – Define the scope of the reconnaissance.
2. **Perform Passive Footprinting** – Collect publicly available information using:
 - WHOIS lookup
 - Search engine analysis (Google Dorks)
 - Company websites
 - LinkedIn and social media
 - DNS information

3. **Analyze the Collected Information** – Identify domains, IP addresses, subdomains, employees, and network details.
4. **Perform Active Scanning** – Conduct ping sweeps, port scanning, DNS enumeration, and service detection only after passive reconnaissance is complete.
5. **Document and Verify Findings** – Validate the collected information and prepare the final intelligence report.

Following this sequence ensures that reconnaissance is systematic, accurate, and less likely to be detected, improving the effectiveness of the footprinting process.

4. A student performs WHOIS lookup but only records domain name and registrar details, ignoring expiry date, administrative contacts, and name servers. Identify the missing components and explain how these fields provide useful security insights. Suggest how proper WHOIS analysis can reveal vulnerabilities.

The student's mistake is recording only the domain name and registrar information while ignoring other important WHOIS fields. A complete WHOIS analysis includes several additional components that provide valuable information for security assessment and footprinting.

Missing Components:

- Domain Expiry Date
- Administrative Contact Information
- Technical Contact Information
- Registrant Information (if publicly available)
- Name Servers (DNS Servers)
- Creation Date
- Last Updated Date
- Domain Status

Security Insights Provided by These Fields:

- **Expiry Date:** Indicates when the domain registration expires. An expired or soon-to-expire domain may be vulnerable to domain hijacking if not renewed.
- **Administrative Contact:** Provides details of the person or organization responsible for managing the domain, which can help identify key personnel during security assessments.
- **Technical Contact:** Identifies the technical administrator responsible for domain configuration and maintenance.
- **Name Servers:** Reveals the DNS servers managing the domain, helping analysts understand the DNS infrastructure and identify possible DNS misconfigurations.

- **Creation and Updated Dates:** Show the age of the domain and recent modifications, which can help detect newly registered or suspicious domains.
- **Domain Status:** Indicates whether the domain is active, locked, or under transfer, helping assess the domain's security configuration.

How Proper WHOIS Analysis Can Reveal Vulnerabilities:

A thorough WHOIS analysis can identify potential security weaknesses, such as:

- Domains that are close to expiration and susceptible to hijacking.
- Misconfigured or outdated DNS servers.
- Publicly exposed administrative or technical contact information that could be targeted in social engineering attacks.
- Recently registered or modified domains that may indicate suspicious activity.
- Weak domain management practices that could increase security risks.

By examining **all WHOIS fields**, security analysts gain a more comprehensive understanding of the target's domain infrastructure, enabling them to identify potential vulnerabilities and improve the accuracy of footprinting and reconnaissance.

5. A student downloads publicly available documents from an organization's website but focuses only on the visible content while ignoring embedded metadata. As a result, important details such as author names, software versions, and internal usernames are missed. Identify the mistake in the student's footprinting process and explain how metadata can provide valuable intelligence. Describe the correct method to extract metadata using appropriate tools. Suggest how this information could be used ethically to identify potential security risks.

The student's mistake is analyzing only the visible content of the downloaded documents while ignoring the embedded metadata. Metadata often contains hidden information that is not visible to users but can provide valuable intelligence during footprinting.

How Metadata Provides Valuable Intelligence:

Metadata can reveal important information such as:

- **Author names** of the document.
- **Software and application versions** used to create or edit the file.
- **Internal usernames** or computer names.
- **Document creation and modification dates.**
- **Company or organization details.**
- **File paths and document properties.**

This information helps security analysts understand the organization's environment, identify technologies in use, and detect potential security weaknesses.

Correct Method to Extract Metadata:

The metadata can be extracted using specialized metadata analysis tools such as:

- **ExifTool**
- **FOCA (Fingerprinting Organizations with Collected Archives)**
- **Metagoofil**
- **pdftinfo** (for PDF documents)
- **Microsoft Office Document Inspector** (for Office files)

Steps:

1. Download publicly available documents from the organization's website.
2. Open the document using a metadata extraction tool.
3. Extract and review metadata fields such as author, software version, creation date, modification date, and document properties.
4. Analyze the extracted information for security assessment and document the findings.

Ethical Use of Metadata Information:

The extracted metadata should only be used for **authorized security assessments** to identify and report potential security risks, such as:

- Outdated software versions that may contain known vulnerabilities.
- Exposure of internal usernames or computer names that could aid attackers.
- Disclosure of unnecessary document properties or file paths.
- Information leakage through hidden metadata.

The findings should be **reported responsibly to the organization** so that exposed metadata can be removed, sensitive information can be protected, and overall security can be improved.

Code of Conduct:

I 192325076 – SHAM S. certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Error Identification	20%				
Root Cause Analysis	30%				
Correction & Justification	25%				
Application of Concepts	15%				
Clarity & Presentation	10%				

Criteria	Error Identification (2)	Root Cause Analysis (3)	Correction & Justification (2)	Application of Concepts (2)	Clarity & Presentation (1)	Total
Q1						
Q2						
Q3						
Q4						
Q5						